

Full provenance travels with every item — source passages · exact prompt · model version · expert ratings · pilot statistics

(a) RAG drafting

retrieve top-k passages
per topic × Bloom level
pinned LLM drafts items:
MCQ · code-tracing ·
code-writing
bilingual (Bangla + English)
pool $\approx 3 \times$ required size

(b) Automated screening

groundedness — key
derivable from sources
format & single-key validity
near-duplicate removal
language check, Bn & En

*filter only —
certifies nothing*

(c) Expert certification

primary certification
panel of 6–8 HSC ICT
teachers & examiners
5 criteria · 4-point scale
unlabelled board-exam
anchors interleaved

GATE
 $I-CVI \geq .78$
 $S-CVI/Ave \geq .90$ · κ reported

discard

(d) Psychometric pilot

secondary certification
pilot cohort $n \approx 15\text{--}20$
(outside main sample)
classical item analysis
+ distractor analysis

GATE
 $.30 \leq P \leq .90$ · $D \geq .20$
clarity $\geq 4.0 / 5$

discard

(e) Certified item bank

frozen & versioned before
data collection begins
item + source passages +
ratings + pilot statistics
released with materials

Study instruments

parallel forms from the
frozen bank only: pre ·
post · transfer · withdrawal

revise & re-certify

If certification yield falls short of the required bank size,
expert-authored items fill the gap; shortfall reported (V1, §3).